



MALLA REDDY UNIVERSITY

(As per Telangana Private Universities Act No. 13 of 2020 &
G.O.Ms.No. 14, Higher Education (UE) Department)

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SCHOOL OF ENGINEERING

DEPARTMENT OF AI & ML 2023-27 (IV Year I Semester)

MR23-1CS0273 Major Project Phase-I

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Name of the Guide	Vinay Simha Reddy	
Project Title	tAI-Agent Based Intelligent Predictive Maintenance System Using Machine Sound Analysis	
Batch Number	11	
Batch Student Details	Roll No	Student Name
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Abstract Work	Predictive maintenance has become an essential approach in modern industries to reduce unexpected machine failures, minimize maintenance costs, and improve operational efficiency. Traditional maintenance methods, such as scheduled inspections and reactive repairs, often fail to identify early signs of machine degradation, leading to unplanned downtime and reduced productivity. This project, "AI-Agent Based Intelligent Predictive Maintenance System Using Machine Sound Analysis," proposes an intelligent solution that utilizes machine sound recordings and Artificial Intelligence (AI) to monitor machine health in real time. The system captures acoustic signals using microphones and processes the audio by extracting features such as Mel-Frequency Cepstral Coefficients (MFCCs) and spectrograms. Machine Learning and Deep Learning models, including Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, are employed to detect abnormal sounds, classify machine faults, and predict the Remaining Useful Life (RUL) of industrial equipment. Based on these predictions, autonomous AI agents analyze machine conditions, determine maintenance priorities, automatically schedule maintenance activities, and send real-time alerts through web browsers and Telegram. An interactive dashboard provides fault classification, machine health status, RUL predictions, maintenance history, and analytical reports. A feedback mechanism enables technicians to validate predictions, allowing continuous model improvement. The proposed system offers a scalable, cost-effective, and intelligent predictive maintenance solution that enhances equipment reliability, reduces downtime, optimizes maintenance planning, and supports Industry 4.0 smart manufacturing environments.	

Project Guide

Project Coordinator

DEAN (AI-ML)